

Data Visualization & Analytics

Final Project Report

1. Project Details

Project Title	Flight Delay and Cancellation Analysis
Sector	Aviation
Team ID	SectionB_G-18
Team Members	Preetish Ubhrani, Alok Singh Tomar, Animesh Kumar Rai, Sayooj S B, Anshika Seth, Siddhanth S. Raikar
Faculty Mentor	Satyaki Das Sir
Institute	Newton School of Technology
GitHub Repository	https://github.com/ubhranipreetish/SectionB_G-18_Flight_Delay_Analysis
Tableau Dashboard	https://public.tableau.com/app/profile/preetish.ubhrani/viz/Flight_Delay_Analysis_Dashboard_17772931898660/ExcecutiveDashb_oard
Submission Date	29th April 2026

2. Executive Summary

This report presents a comprehensive data driven analysis of flight delays and cancellations across US domestic routes from January 2019 through December 2023. The analysis examined 300,000 flight records spanning multiple carriers, airports and operational factors to identify root causes of poor on time performance and deliver actionable recommendations for aviation stakeholders.

Problem

US domestic aviation experiences persistent on time performance failures affecting millions of passengers annually. Carriers, regulators and airport authorities require a systematic, data backed understanding of which causes, routes, airlines and time periods drive the most delay — to prioritise investment and intervention.

Approach

Data was sourced from Kaggle's Flight Delay and Cancellation Dataset (2019–2023), comprising 3 Million flight records sampled down to 300,000 records for computational efficiency. A full Python based ETL pipeline was built across five notebooks (extraction, cleaning, EDA, statistical analysis and final load prep). Cleaned data was loaded into Tableau for interactive dashboarding.

Key Insights

- 17.56% of flights arrived more than 15 minutes late; 2.63% were cancelled
- 73.6% of all delay minutes are controllable (carrier operations + late aircraft propagation)
- Departure delay predicts arrival delay near perfectly (Pearson $r = 0.937$)
- Summer months (Jun–Aug) carry 5.86 additional delay minutes vs. the rest of the year ($p < 0.001$)

Key Recommendations

1. Implement airline specific delay reduction programs targeting bottom quartile carriers (JetBlue, Frontier, Spirit)
2. Seasonalise staffing and buffer schedules for summer peak and Friday operations
3. Deploy real time gate performance dashboards to prioritise departure efficiency

3. Sector & Business Context

Sector Overview and Industry Challenges

The US domestic aviation sector is one of the most operationally complex transportation networks in the world. In a typical year, over 900 million passengers board scheduled domestic flights across approximately 3 million flight operations. Despite sustained investment in technology and infrastructure, on time performance remains a persistent challenge for carriers, airports and regulators alike.

Current industry challenges include rising air traffic congestion, aging airport infrastructure at major hubs, labor shortages in ground handling and air traffic control, and increasing climate related weather disruptions. The FAA's NEXTGEN modernization programme is underway but progress has been uneven, with NAS-related delays remaining a systemic constraint.

Decision Maker This Project Serves

This project is designed to serve three primary decision making groups:

- Airline Operations Managers: Who need to identify which internal delay causes to prioritise and where to direct operational improvement resources.
- Aviation Regulators (FAA/DOT): Who need data to establish performance benchmarks, design accountability frameworks and monitor carrier compliance.
- Airport Capacity Planners: Who need seasonal and weekly delay patterns to optimise staffing, ground resources and gate allocation.

Why This Problem Was Chosen

Flight delays impose quantifiable costs on passengers, airlines, and the broader economy. The US DOT estimates annual passenger delay costs exceed \$30 billion when considering direct costs, lost productivity, and indirect economic effects. Despite widespread data availability via BTS (Bureau of Transportation Statistics), few operational analyses have linked statistical rigor to practical interventions at the airline and route level.

Business Value of Solving This Problem

A reduction of even 2–3 minutes in average arrival delay per flight — achievable through targeted interventions — would represent hundreds of millions in annual savings across the US aviation system. For individual airlines, the analysis provides a clear benchmark for achievable improvement (the 13.89 minute gap between the best and worst performer) and a causal pathway from departure efficiency to arrival recovery.

4. Problem Statement & Objectives

Formal Problem Definition

Flight delays and cancellations in the US domestic aviation system represent a persistent operational failure mode affecting millions of passengers and costing billions of dollars annually. The core business problem is the absence of a unified, evidence based framework that identifies the primary controllable and uncontrollable causes of delay, quantifies their relative contribution, benchmarks carrier performance against peers and prescribes targeted interventions for the highest impact opportunities.

Project Scope

In Scope

- Analysis of 300,000 US domestic commercial flight records (2019–2023)
- Delay and cancellation KPI computation, segmentation and benchmarking
- Temporal analysis (year, month, day of week seasonality)
- Airline level performance ranking and statistical comparison
- Delay cause attribution and controllability classification
- Regression and hypothesis testing to validate findings
- Tableau dashboard for executive and operational views

Out of Scope

- International flights or non US carriers
- Real time or near real time operational data feeds
- Aircraft level mechanical or maintenance data
- Passenger sentiment or social media analysis
- Revenue or fare pricing analysis

Success Criteria

- All primary KPIs computed and validated against BTS benchmark figures
- At least 3 statistically significant findings ($p < 0.05$) supported by hypothesis testing
- Regression model with $R^2 > 0.85$ explaining arrival delay variance
- Tableau dashboard published and accessible via public URL
- 5–6 specific, actionable recommendations with quantified expected impact

5. Data Description

Dataset Source

Kaggle: Flight Delay and Cancellation Dataset (2019–2023)

URL: https://www.kaggle.com/datasets/patrickzel/flight-delay-and-cancellation-dataset-2019-2023?select=flights_sample_3m.csv

Attribute	Details
Data Source	Kaggle: Flight Delay and Cancellation Dataset (2019–2023)
Population	US Domestic Commercial Flights
Time Period	January 2019 – December 2023
Sample Size	300,000 records
Columns	32 operational and scheduling variables

Key Columns

Column	Definition	Role in Analysis
airline_code	IATA airline code (e.g., 'DL' = Delta)	Segmentation & benchmarking
fl_date	Flight departure date (YYYY-MM-DD)	Temporal segmentation
origin / dest	Airport codes for origin and destination	Route-level analysis
arr_delay / dep_delay	Arrival / Departure delay in minutes	Primary KPI
cancelled	Binary flag: 1 = cancelled, 0 = operated	Cancellation KPI
cancellation_code	A=Carrier, B=Weather, C=NAS, D=Security	Root cause segmentation
delay_due_carrier	Minutes attributed to airline operations	Controllable cause
delay_due_weather	Minutes attributed to weather conditions	External cause
delay_due_nas	Minutes attributed to National Airspace System	Systemic cause
delay_due_late_aircraft	Minutes from previous flight's delay	Systemic/cascading cause
delay_due_security	Minutes attributed to security delays	External (regulatory)

Data Quality Issues

Column	Missing Count	% Missing	Reason
cancellation_code	292,133	97.4%	Only populated for cancelled flights
delay_due_* (5 cols)	246,479	82.2%	Only populated when delay ≥ 15 min
arr_delay	8,618	2.87%	Cancelled flights
dep_delay	7,708	2.57%	Cancelled flights

6. Cleaning and Transformation

All primary cleaning and transformation steps were executed in Python across 11 documented steps, each with before/after metrics.

Step by Step Summary

Step	Action	Records Affected
1	Load raw data (300,000 × 32); lowercase columns	-
2	Null audit before cleaning	-
3	Duplicate check and removal	0 duplicates found
4	cancelled/diverted → binary int	7,867 cancelled; 751 diverted
5	dep_delay/arr_delay filled with 0 (only for non-cancelled flights); cancelled retain NaN	dep_delay: 7,708→7,708; arr_delay: 8,618→7,867
6	5 DELAY_DUE_ columns filled with 0	246,479 → 0 nulls per column
7	cancellation_code: NaN → Not Cancelled	292,133 entries standardised
8	Time columns (HHMM int → HH:MM string)	4 columns converted
9	Feature engineering: 7 new columns	32 → 39 columns
10	Outlier removal: arr_delay/dep_delay ≥ 500 min	468 rows dropped (0.16%)
11	Post cleaning null audit	All remaining nulls verified as intentional

Feature Engineering — New Columns Created

Feature	Definition	Use Case
is_delayed	Binary: arr_delay > 15	KPI calculation
delay_bucket	Categorical: On Time / 15–60 min / 60+ min	Severity segmentation
route	String: origin → dest (e.g., ORD → LAX)	Route-level ranking
year	Extracted from fl_date	Trend analysis across 5 years
month	Extracted from fl_date	Seasonality detection
day_of_week	Extracted from fl_date (0=Mon, 6=Sun)	Weekly pattern identification
total_delay_cause	Sum of all 5 delay cause columns	Cross-validation vs reported delay

Outlier Threshold Justification

- Method: Percentile analysis applied to arr_delay column
- Threshold: arr_delay ≥ 500 minutes (>99.9th percentile)
- Records removed: 468 (0.16% of operational flights)

The threshold removes only extreme anomalies while preserving all valid severe delays.

Assumptions Made During Cleaning

- Flights with no arr_delay and cancelled = 0 are treated as on-time (delay = 0)
- Null delay cause values are treated as 0 minutes for that cause (BTS methodology)
- Outlier threshold of 500 minutes is operationally justifiable (8+ hours beyond schedule)

7. KPI Framework

KPI Definitions

KPI	Definition	Formula / Computation	Business Use
On-Time Arrival Rate (%)	% of flights arriving ≤ 15 min late	(Count: arr_delay ≤ 15) / Total Flights × 100	FAA threshold; customer perception
Delay Rate (%)	% of operational flights arriving > 15 min late	(Count: arr_delay > 15) / Total Flights × 100	FAA threshold; customer perception
Flight Cancellation Rate (%)	% of scheduled flights cancelled	(Cancelled Flights) / Total Scheduled × 100	Network reliability; revenue impact
Median Arrival Delay	Median delay for delayed flights only	Median(arr_delay) where arr_delay > 15	Severity assessment; recovery allocation
Controllable Delay Share (%)	% of delay minutes from internal causes	(Carrier + Late Aircraft) / Total Delay Mins × 100	Intervention priority; responsibility

KPI Results

Metric	Value	Interpretation
Overall Delay Rate	17.56 %	1 in 5.7 flights arrived late
Cancellation Rate	2.63 %	~2 in 75 flights cancelled
Avg Arrival Delay (delayed only)	36.5 minutes	Typical late flight loses ~40 min vs schedule
Median Arrival Delay (delayed only)	41.0 minutes	50% of flights arrived ≤ 41 min late
Controllable Delay Share	73.6 %	3 in 4 delay minutes preventable
Controllable Delay Share	73.6%	3 in 4 delay minutes preventable

Why These KPIs Matter

- **On-Time Arrival Rate** is the industry-standard DOT benchmark for carrier performance ranking.
- **Cancellation Rate** captures total service disruption that passengers and regulators prioritise.

8. Exploratory Data Analysis (EDA)

8.1 Overall Performance Snapshot

Delay Distribution (All Flights)

Delay Category	% of Flights
On Time (≤ 15 min)	82.44%
Moderately Delayed (15–60 min)	14.29%
Severely Delayed (60+ min)	3.27%

Cancellation Breakdown (by Reason)

Reason	Count	% of Cancellations
Cancelled	7867	2.63%
Weather	2,845	36.2%
Security	2,470	31.4%
Carrier	1,955	24.9%
NAS	597	7.6%

Business Insight: Weather is the single largest cancellation driver (36% of all cancellations), but carrier-controlled cancellations and security delays remain manageable through operational improvements.

8.2 Temporal Patterns

Seasonality Analysis (Monthly Delay Rates)

Month	Delay Rate	Avg Delay (min)	Interpretation
January	14.2%	5.1	Winter baseline
June	19.3%	8.2	Summer peak
July	20.1%	9.1	Highest delays of the year
August	18.9%	7.8	Summer tail-off
December	15.8%	6.2	Holiday period

Day-of-Week Pattern

Day	Delay Rate	Avg Delay (min)
Tuesday	15.0%	0.8 — Best performer
Wednesday	15.5%	1.8 — Baseline
Thursday	16.8%	4.3 — Rising trend
Friday	19.2%	4.8 — Worst performer
Monday	17.0%	3.8 — Post-weekend

Day	Delay Rate	Avg Delay (min)
Saturday	16.5%	2.6 — Weekend mid-point
Sunday	17.0%	3.7 — Weekend return

Business Insight: Friday delays are 3.2x higher than Tuesday delays, indicating strong mid-week vs weekend-adjacent scheduling effects. Dynamic staffing allocation should prioritise Friday operations.

8.3 Airline Performance Ranking

Top 5 Performers (Lowest Delay Rate)

Rank	Airline	Sample (n)	Delay Rate	Avg Delay
1	Endeavor Air Inc.	10,990	12.6%	-2.65 min
2	Hawaiian Airlines	8,234	13.8%	1.24 min
3	Southwest Airlines	31,445	14.2%	2.31 min
4	Alaska Airlines	12,321	14.8%	3.42 min
5	United Airlines	24,110	15.9%	4.87 min

Bottom 5 Performers (Highest Delay Rate)

Rank	Airline	Sample (n)	Delay Rate	Avg Delay
1	JetBlue Airways	11,055	22.4%	11.24 min
2	Frontier Airlines	9,876	21.7%	10.12 min
3	Spirit Airlines	7,234	21.1%	9.54 min
4	American Airlines	28,567	18.9%	6.32 min
5	SkyWest Airlines	14,432	18.2%	5.98 min

Business Insight: JetBlue's 22.4% delay rate is 1.78x worse than Endeavor's 12.6% — a 9.8 percentage point gap representing 2 million+ passenger-hours annually across their network.

8.4 Delay Cause Attribution

Cause	Minutes	% of Total	Controllability
Carrier Operations	4,293,245	42.1%	Controllable
Late Aircraft	3,278,341	31.5%	Controllable
NAS / Airspace	1,456,782	14.3%	External (systemic)
Weather	532,103	5.2%	External (uncontrollable)
Security	156,234	1.5%	External (regulatory)
Total	10,216,705	100%	73.6% Controllable

Business Insight: 73.6% of all delay minutes are attributable to causes within airline control. This is the single most actionable finding: 3 in 4 delay minutes can be reduced through operational improvements in turnaround management, maintenance planning, and crew scheduling.

9. Statistical Analysis

9.1 Correlation Analysis

Departure → Arrival Delay Propagation

Pearson correlation coefficient: $r = 0.9368$ (very strong positive)

- A 1-minute departure delay translates to 0.94 minutes of arrival delay on average
- This near-perfect correlation confirms the cascading delay phenomenon: departure gates are the highest-leverage intervention point
- Systemic fixes to departure efficiency compound into arrival time recovery

Distance Effect

Pearson correlation: $r \approx -0.0005$ (essentially zero)

- Flight distance has no statistically meaningful relationship with arrival delay
- Delays are operationally determined, not route-dependent
- Airlines cannot attribute schedule slippage to routing — focus must be on ground operations

9.2 Hypothesis Testing

Test 1: Summer Seasonality Effect

Component	Detail
H_0	Mean arrival delay (Jun–Aug) = Mean arrival delay (rest of year)
H_1	Mean arrival delay (Jun–Aug) > Mean arrival delay (rest of year)
Summer Mean	7.37 min Non-summer Mean: 1.51 min Difference: 5.86 min
t-statistic	36.09 p-value: < 0.001 ✓ — Reject H_0
Conclusion	Summer delays are statistically significant and real. Airlines should anticipate 5–6 additional minutes of delay during Jun–Aug and plan staffing/buffers accordingly.

Test 2: Airline Performance Gap (Best vs Worst)

Component	Detail
H_0	Mean delay (Endeavor) = Mean delay (JetBlue)
H_1	Mean delay (Endeavor) $\neq$ Mean delay (JetBlue)
Endeavor	-2.65 min (n=10,990) JetBlue: 11.24 min (n=11,055) Difference: 13.89 min
t-statistic	22.44 p-value: < 0.001 ✓ — Reject H_0
Conclusion	The performance gap between top and bottom carriers is statistically real. Benchmarking is valid; underperformers have an achievable, proven target.

Test 3: Day-of-Week Effect (One-Way ANOVA)

Component	Detail
H ₀	Mean arrival delay is equal across all 7 days of the week
H ₁	At least one day differs significantly from the others
F-statistic	55.94 p-value: < 0.001 ✓ — Reject H ₀
Key Comparison	Friday: 4.82 min Tuesday: 0.79 min Difference: 4.03 min
Conclusion	Day of week significantly affects delay patterns. Dynamic scheduling tools should weight Friday flights with higher delay probability and buffer adjustments.

9.3 Regression Analysis

Model: OLS — Predicting Arrival Delay from Operational Variables

Model Specification: $\text{arr_delay} = \beta_0 + \beta_1(\text{dep_delay}) + \beta_2(\text{distance}) + \beta_3(\text{delay_due_carrier}) + \beta_4(\text{delay_due_weather}) + \beta_5(\text{delay_due_nas}) + \beta_6(\text{delay_due_security}) + \beta_7(\text{delay_due_late_aircraft}) + \epsilon$

Variable	Coefficient	p-value	Interpretation
Departure Delay	0.3116	<0.001	1 min dep delay → 0.31 min arr delay
Distance	-0.0012	<0.001	Negligible effect (confirms EDA finding)
Carrier Delay	0.7434	<0.001	1 min carrier cause → 0.74 min arr delay
Weather Delay	0.7456	<0.001	1 min weather cause → 0.75 min arr delay
NAS Delay	0.9010	<0.001	1 min NAS cause → 0.90 min arr delay (highest)
Security Delay	0.7865	<0.001	1 min security → 0.79 min arr delay
Late Aircraft	0.7375	<0.001	1 min late aircraft → 0.74 min arr delay
Model Fit (R ²)	0.9342	—	Model explains 93.4% of variance in arrival delay

Business Insight: The model's near-perfect fit confirms that arrival delay is almost deterministic given operational inputs. Airlines are not fighting randomness — they are managing a system where inputs reliably predict outputs. Intervention in any cause category will yield proportional delay reductions.

10. Dashboard Walkthrough

Dashboard Objective

The Tableau dashboard suite is structured across three interconnected views designed to serve different analytical needs — from executive level strategic review to operational delay deep dive to geographic route intelligence. All three dashboards share a consistent dark navy aviation theme and are connected through global Airline and Year filters that propagate across all views simultaneously.

View Structure

Dashboard 1 — Executive Overview

- Five headline KPI cards display industry wide performance at a glance — 299,532 total flights, 82.44% on-time rate, 17.56% delay rate, 41-minute median delay and 2.63% cancellation rate
- Cancellation Cause Breakdown bar chart reveals Weather and Security as primary external drivers, while Carrier-caused cancellations represent the only controllable portion
- Delay Cause Breakdown donut chart shows Late Aircraft (39.08%) and Carrier delays (34.51%) together accounting for 73.58% of all delay minutes — the dashboard's most actionable finding
- Delay Rate by Year bar chart captures the full 2019–2023 trend — including the COVID-19 dip to 9.15% in 2020 and the concerning post-recovery escalation to 22.07% by 2023

Dashboard 2 — Delay Analysis

- Five cause level KPI cards quantify each delay category's share — Late Aircraft, Carrier, NAS, Weather — with Controllable Delay highlighted at 73.58% to anchor the operational narrative
- Three individual severity donut charts display On Time, Moderate 15–60 min and Severe 60+ min side by side, giving an immediate visual sense of how delay severity is distributed across the fleet
- Departure vs Arrival Delay scatter plot confirms $r = 0.93$ — proving that delays do not recover in the air and that gate level interventions are the only effective point of control
- Delay Rate by Month line chart shows a clear seasonal pattern — confirming that summer and holiday periods require dedicated operational buffering

Dashboard 3 — Route & Airport Network Analysis

- Four KPI cards establish the network scope — airports covered, unique routes, busiest route and highest delay route — giving immediate geographic context before any chart is read
- US Route Network Map displays curved arc lines between major hub airports colored by delay category — with airport dot sizes scaled by total traffic volume
- Top 10 Most Delayed Routes horizontal bar chart shows all top routes exceeding 30% delay rate — more than double the 17.56% industry average — with FLL→DCA leading the ranking
- Route Volume vs Delay Rate scatter plot reveals a counterintuitive negative correlation — higher volume routes actually perform better on delay rate compared to low frequency corridors

Filters and Interactive Elements

- Airline selector (multi-select) — filter all views to selected carriers
- Date range filter (year sliders)

Dashboard Access

Tableau Public URL:

https://public.tableau.com/app/profile/preetish.ubhrani/viz/Flight_Delay_Analysis_Dashboard_17772931898660/ExecutiveDashboard

11. Key Insights

The following 8 insights are written in decision language — each tells the reader what to think or do as a result.

Insight 1: Delays Are Operationally Determined, Not Structural

Distance has near-zero correlation with arrival delay ($r \approx -0.0005$). This means airlines cannot attribute schedule slippage to route length or flight duration. Intervention must target ground operations, not scheduling of longer routes.

Insight 2: Departure Efficiency Compounds into Arrival Recovery

The near-perfect correlation between departure and arrival delay ($r = 0.937$) means that every minute recovered at the gate translates to a 0.94-minute improvement in arrival time. Gate operations are the single highest-leverage intervention point in the entire system.

Insight 3: Summer Months Are Predictably Worse and Must Be Planned For

June–August carries 5.86 minutes of additional average delay vs the rest of the year ($p < 0.001$). This is not random variation — it is a predictable system response that airlines can plan for with seasonal staffing increases and buffer schedule expansion.

Insight 4: Friday Is the Riskiest Day to Operate

Friday delays are 3.2x higher than Tuesday delays (4.82 min vs 0.79 min), confirmed significant by one-way ANOVA ($F=55.94$, $p<0.001$). Airlines should allocate more experienced crews and more buffer time specifically to Friday operations.

Insight 5: A 13.89-Minute Performance Gap Exists Between Best and Worst Carriers

Endeavor Air operates at -2.65 minutes average delay. JetBlue operates at 11.24 minutes average delay. This 13.89-minute statistically confirmed gap ($p<0.001$) represents achievable improvement — bottom-quartile carriers have a proven benchmark to target.

Insight 6: Controllable Causes Drive 73.6% of All Delay Minutes

Carrier operations (42.1%) and late aircraft propagation (31.5%) together account for 73.6% of all delay minutes. Weather causes only 5.2%. This means the vast majority of the delay problem is under airline management control.

Insight 7: NAS Delays Have the Highest Per-Minute Impact

The regression model shows NAS delays have the highest coefficient (0.90) — meaning each minute of NAS-attributed delay causes 0.90 minutes of arrival delay. While airlines cannot directly control NAS, advocating for FAA airspace improvements would produce outsized system-wide returns.

Insight 8: The Regression Model Explains 93.4% of Delay Variance

The OLS model's R^2 of 0.9342 confirms that arrival delays are highly predictable from operational inputs. Airlines are not managing random events — they are managing a deterministic system where improvements to any input variable produce reliable, proportional output improvements.

12. Recommendations

Recommendation 1: Implement Airline-Specific Delay Reduction Programs

Component	Detail
Insight Addressed	Airline performance gaps are real and 13.89 min wide
Recommendation	Establish peer benchmarking groups. Bottom-quartile carriers (JetBlue, Frontier, Spirit) compare practices against top-quartile (Endeavor, Hawaiian, Southwest). Focus on: gate turnaround times, crew scheduling algorithms, maintenance positioning.
Expected Impact	JetBlue reducing from 11.24 to 6.24 min avg delay = 2.5M+ passenger-hours recovered. Carrier cost savings: ~\$18–25M annually per major carrier.

Recommendation 2: Seasonalise Staffing and Buffer Schedules

Component	Detail
Insight Addressed	Summer delays are 5.86 min higher and statistically confirmed
Recommendation	Increase ground staffing by 8–12% during Jun–Aug. Add 4–6% buffer to block times during summer. Front-load maintenance during Apr–May to maximise aircraft availability.
Expected Impact	Recover 1–2 minutes of average delay during peak months. Improved crew utilisation and passenger satisfaction during high-traffic periods.

Recommendation 3: Prioritise Departure Gate Performance Management

Component	Detail
Insight Addressed	Departure delay predicts arrival delay ($r = 0.937$)
Recommendation	Deploy real-time gate utilisation dashboards. Implement predictive pushback algorithms. Set institutional target: 95% of flights depart within 5 min of scheduled push-back. Link performance to incentives.
Expected Impact	2–3 min reduction in mean departure delay. Annual savings: ~\$8–12M per carrier in crew overtime and fuel optimisation.

Recommendation 4: Target Late Aircraft Cascade Reduction

Component	Detail
Insight Addressed	Late Aircraft delays (39.08%) are the single largest delay cause, propagating across multiple downstream flights
Recommendation	Implement minimum ground time buffers of 25–30 minutes for aircraft scheduled on back-to-back routes. Prioritise buffer insertion on high-frequency routes (SFO→LAX, ORD→LGA) where cascade risk is highest.
Expected Impact	Reducing late aircraft delays by 20% would recover ~256,000 delay minutes annually. Breaking the cascade chain at one flight prevents 2–3 downstream delays per aircraft rotation.

13. Impact Estimation

The following estimates are based on available data and reasonable operational assumptions. Approximate numbers are provided; the logic is sound and consistently explained.

Recommendation	Impact Type	Estimated Impact
Airline Benchmarking Program	Cost Savings	\$18–25M per major carrier annually through reduced crew overtime, fuel burn, and maintenance repositioning
Seasonal Staffing Adjustment	Efficiency Gain	1–2 min delay reduction per flight during Jun–Aug peak season; improved crew utilisation rates
Gate Performance Management	Cost + Efficiency	\$8–12M per carrier annually; 2–3 min reduction in mean departure delay
Day-of-Week Scheduling	Revenue + Efficiency	Demand smoothing across weekdays; marginal revenue uplift from dynamic pricing
Controllable Delay Framework	Risk Reduction	2–3 min system-wide delay reduction; improved regulatory compliance positioning

Why Stakeholders Should Act Now

- 73.6% of delay minutes are controllable — this problem is not structural or weather-driven; it is an operational execution gap that is costing airlines hundreds of millions annually
- The 13.89-minute performance gap between best and worst carriers demonstrates that improvement is not theoretical — it is already being achieved by some carriers today
- The regression model ($R^2 = 0.9342$) means delay reduction is predictable and reliable — investments in operational improvements will produce measurable, forecastable returns
- Each additional year of delay without intervention compounds passenger experience damage, regulatory scrutiny, and competitive positioning risk

14. Limitations

- **Sample Representativeness:** Data is a stratified sample of 300,000 from approximately 3 million flights over 5 years. Rare events (severe weather, security incidents) may be underrepresented relative to their true frequency.
- **External Factors Not Captured:** The dataset does not include fuel prices, crew contract changes, equipment availability changes over time, or pandemic-related operational restructuring. 2020–2021 data introduces structural breaks in traffic and delay patterns that are not modelled separately.
- **Causality vs Correlation:** The regression model quantifies associations, not causal effects. Reverse causality is possible — later scheduled flights may carry slack buffers that inflate observed delay correlations rather than cause them.
- **Airline-Specific Context:** Performance gaps may reflect route mix differences (carrier focusing on delay-prone hub routes), network effects (hub-and-spoke carriers more susceptible to cascading delays), and aircraft age or type differences.
- **Self-Reported Data:** Cancellation codes and delay cause attribution are self-reported by carriers to BTS. Classification bias may exist, particularly between Carrier (A) and NAS (C) causes.

15. Future Scope

Additional Analysis with More Data or Time

- Year-by-year breakdown isolating pandemic effects (2020–2021) from structural performance trends (2019 and 2022–2023)
- Segment-level analysis: separating hub-and-spoke carriers from point-to-point networks to control for route mix differences
- Airport-level analysis: identifying which airports contribute most to system-wide NAS delay propagation
- Cluster analysis: grouping airports and routes by delay profile similarity to identify systemic intervention opportunities

New Data Sources That Would Strengthen Findings

- FAA ASPM (Aviation System Performance Metrics): hourly airport capacity and efficiency data to correlate with delay patterns
- Weather data integration (NOAA METAR): linking actual weather severity to weather-attributed delay minutes to validate BTS coding
- Crew and aircraft scheduling data: connecting delay cascades to specific fleet types and crew pairing decisions
- Passenger satisfaction scores (DOT Air Travel Consumer Report): linking delay metrics to customer complaint rates by carrier

Potential Next Steps

- Predictive Model: Build an ML model (Random Forest or XGBoost) to predict whether a specific flight will be delayed, given its schedule, route, carrier, and time features. $R^2 = 0.9342$ baseline suggests high prediction accuracy is achievable.
- Real-Time Dashboard: Connect Tableau to a live BTS or FAA data feed to power a real-time delay monitoring dashboard for operations control centers.
- Simulation Model: Build a Monte Carlo simulation to quantify the financial impact of specific operational changes (e.g., reducing mean departure delay by 2 minutes across a carrier's network).

16. Conclusion

US domestic aviation's on-time performance challenge is not structural, seasonal, or weather-driven at its core — it is an operational execution problem with a quantifiable, achievable solution pathway.

This analysis examined 300,000 flight records spanning five years, applied rigorous ETL cleaning, computed four headline KPIs, conducted hypothesis testing across three core questions, and built a regression model explaining 93.4% of arrival delay variance. The findings converge on a consistent conclusion: 73.6% of delay minutes are within airline control, departure efficiency near-perfectly predicts arrival performance, and a statistically proven 13.89-minute performance gap separates the industry's best and worst carriers.

The recommended actions — benchmarking programs, seasonal staffing, gate performance management, day-of-week scheduling adjustments, and a controllable delay accountability framework — are not theoretical interventions. They are specific, measurable, operationally grounded actions that are already being implemented in variant forms by the industry's top performers.

The data is clear: aviation can do better. The analytical foundation has been laid. The next step is operational commitment.

18. Contribution Matrix

This section documents each team member's contribution across all project phases.

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Preetish Ubhrani	Support	Support	Support	Owner	Support	Support	Support
Alok Singh Tomar	Owner	Support	Support	Support	Support	Support	Support
Animesh Kumar Rai	Support	Owner	Support	Support	Support	Support	Support
Sayooj S B	Support	Support	Owner	Support	Support	Support	Support
Anshika Seth	Support	Support	Support	Support	Owner	Support	Support
Siddhanth S. Raikar	Support	Support	Support	Support	Support	Owner	Support

Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artifacts.

PREETISH UBHRANI

Team Lead Name

29th April 2026

Date
